DOLBY

Dolby Media Decoder

User's manual

Version 2.5
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Contents

1 Introduction to the Dolby Media Decoder user's manual

This manual describes how to use the Dolby Media Decoder.

- About this documentation
- Abbreviated speaker notations
- Contacting Dolby

1.1 About this documentation

This manual is for content professionals who need to decode and monitor Dolby audio bitstreams that were created with Dolby Media Encoder from standard channel-based audio or a Dolby Atmos master file.

Dolby Media Decoder supports the following encoded formats:

- Dolby TrueHD bitstream with Dolby Atmos
- Dolby Digital Plus bitstream with Dolby Atmos
- Dolby TrueHD bitstream
- Dolby Digital Plus bitstream
- Dolby Digital bitstream
- MLP Lossless bitstream

Dolby Media Decoder supports monitoring and auditioning of consumer listening modes, including downmixing and compression modes.

1.2 Abbreviated speaker notations

Dolby documentation uses specific names and abbreviations to describe speakers or speaker channels. Dolby notation can differ from the names and abbreviations used by other organizations and companies.

Note: This information is provided for reference only. Not all speaker outputs identified in the table are supported by Dolby Media Producer Suite.

Table 1: Dolby speaker notation correspondences

Dolby notation		CEA-861	ITU-R BS.2159-5 (Type B 10.2 channel)	SMPTE 2036-2 (22.2 channel)	IEC 62574 (30.2 channel)
L	Left	FL	L	FL	FL
R	Right	FR	R	FR	FR
C	Center	FC	C	FC	FC
Ls	Left surround	RL	LS	—	LS, LSd
Rs	Right surround	RR	RS	—	RS, RSd
Lrs	Left rear surround Left back	RLC	LB	BL	BL

Table 1: Dolby speaker notation correspondences (continued)

Dolby notation		CEA-861	ITU-R BS.2159-5 (Type B 10.2 channel)	SMPTE 2036-2 (22.2 channel)	IEC 62574 (30.2 channel)
Rrs	Right rear surround Right back	RRC	RB	BR	BR
Cs	Center surround	RC	—	BC	BC
Lw	Left wide	FLW	—	—	FLW
Rw	Right wide	FRW	—	—	FRW
Lc	Left center	FLC	—	FLC	FLC
Rc	Right center	FRC	—	FRC	FRC
Ls1	Left surround 1	—	—	SiL	SiL
Rs1	Right surround 1	—	—	SiR	SiR
Lfh Lvh	Left front height	FLH	LH	TpFL	TpFL
Rfh Rvh	Right front height	FRH	RH	TpFR	TpFR
Ltm Lts	Left top middle Left top surround	—	—	TpSiL	TpSiL
Rtm Rts	Right top middle Right top surround	—	—	TpSiR	TpSiR
Ltf	Left top front	—	—	—	—
Rtf	Right top front	—	—	—	—
Ltr	Left top rear	—	—	—	TpLS
Rtr	Right top rear	—	—	—	TpRS
Lrh	Left rear height	—	—	TpBL	TpBL
Rrh	Right rear height	—	—	TpBR	TpBR
LFE LFE1	Low-frequency effects 1	LFE	LFE1	LFE1	LFE1
LFE2	Low-frequency effects 2	—	LFE2	LFE2	LFE2
Lsc	Left screen	—	—	—	—
Rsc	Right screen	—	—	—	—
Ls2	Left surround 2	—	—	—	—
Rs2	Right surround 2	—	—	—	—
Lcs	Left center surround	—	—	—	—
Rcs	Right center surround	—	—	—	—
Ch	Center height	FCH	—	TpFC	TpFC
Ts	Top surround	TC	—	TpC	TpC
Lbin	Left binaural	—	—	—	—
Rbin	Right binaural	—	—	—	—
Lo	Left only	—	—	—	—
Ro	Right only	—	—	—	—
Lsd	Left surround direct	—	—	—	—

Table 1: Dolby speaker notation correspondences (continued)

Dolby notation		CEA-861	ITU-R BS.2159-5 (Type B 10.2 channel)	SMPTE 2036-2 (22.2 channel)	IEC 62574 (30.2 channel)
Rsd	Right surround direct	—	—	—	—
Le	Left Dolby Atmos enabled	—	—	—	—
Re	Right Dolby Atmos enabled	—	—	—	—
Lse	Left surround Dolby Atmos enabled	—	—	—	—
Rse	Right surround Dolby Atmos enabled	—	—	—	—
Lrse	Left rear surround Dolby Atmos enabled	—	—	—	—
Rrse	Right rear surround Dolby Atmos enabled	—	—	—	—
—		—	—	BtFC	BtFC
—		—	—	BtFL	BtFL
—		—	—	BtFR	BtFR
—		—	CH	TpBC	TpBC

1.3 Contacting Dolby

You can contact Dolby regarding this product and its supporting documentation.

For technical or product-related questions, contact DMPsupport@dolby.com.

For comments about this documentation, contact documentation@dolby.com.

2 Getting started with Dolby Media Decoder

The Dolby Media Decoder application is part of the Dolby Media Producer Suite. It is installed when the full suite is installed. Refer to the *Dolby Media Producer Suite release notes* for system requirements and installation steps.

2.1 Starting Dolby Media Decoder

The Dolby Media Decoder application provides access to all Dolby Media Decoder user-facing controls.

About this task

The Dolby Media Decoder Client application icon is labeled Dolby Media Decoder.

Procedure

To launch the application, do one of the following actions:

- Click the Dolby Media Decoder application icon.

- Double-click the Dolby Media Decoder application file (located in /Applications/Dolby/Dolby Media Producer/).

Results

The Dolby Media Decoder main window appears.

2.2 Quitting Dolby Media Decoder

You can quit a Dolby Media Decoder session by using the Dolby Media Decoder menu or icon.

Procedure

To quit the application, do one of the following actions:

- On the Dolby Media Decoder menu bar, select **Dolby Media Decoder** > **Quit Dolby Media Decoder**.
- Right-click the Dolby Media Decoder icon, and select **Quit**.

- Click the X button in the upper-left corner of the Dolby Media Decoder window.

2.3 Checking the software version number

You can check the version number of the Dolby Media Decoder, including the version number of the parent Dolby Media Producer Suite.

Procedure

To check the software version number, do one of the following actions:

- On the menu bar, select **Dolby Media Decoder** > **About Dolby Media Decoder**.
- Click the Dolby logo at the top-left corner of a Dolby Media Decoder window.

3 Dolby Media Decoder setup

When Dolby Media Decoder is installed, it is initially configured to use the built-in line-level outputs to decode two-channel audio to the speakers attached to the local machine. If you wish to use external hardware or more than two speakers, you must set up a new speaker configuration for the decoder.

- Setting up Dolby Media Decoder
- Setting speaker configurations
- Creating listening modes

3.1 Setting up Dolby Media Decoder

The Dolby Media Decoder allows you to be set according to your speaker configuration.

Procedure

1. Set up your audio output, either internal or external.
2. Set up your speaker configurations.
3. Test the results.

   Note: In addition to a proper application setup, a carefully calibrated listening environment is crucial to proper content evaluation. You need to do this only once for each mode; however, we recommend that you perform a quick check before each session to reconfirm the correct settings.
4. Optional: Configure custom compression options.
5. Optional: When you are satisfied with your monitoring configurations, save them as a decoder configuration files.

Related information

Internal and external audio hardware on page 17
Setting speaker configurations on page 10
COMPRESSION section on page 32
Saving configuration files on page 23

3.2 Setting speaker configurations

Use the **Speaker Settings** page of the Preferences window to configure and manage as many as eight distinct user-named listening modes. Each mode can accommodate 2 to 16 channels.

About this task

By default, when you first launch the decoder, the speaker configuration is set to stereo using the built-in output of your computer, so that the decoder is ready for immediate use.

Figure 1: Default speaker configuration on the Speaker Settings page

You can assign all user modes to the same I/O device to feed the same speakers, or you can assign any number of modes to different I/O devices to feed different sets of speakers. For example, you might set up three modes: a Dolby Atmos mode that feeds a set of 16 speakers to audition bitstreams carrying Dolby Atmos audio and metadata, a 5.1-channel mode that feeds a set of just six large speakers using the same external audio I/O, and a two-channel mode using a separate external I/O that feeds a pair of small stereo speakers.

Each mode has separate settings for level, delay, and bass management. Each mode also has a custom name assigned by you. When modes have been saved on the **Settings** page in the **Preferences** window, they appear in the **LISTENING** section of the monitor panel.

Procedure

1. To open the **Settings** page, select **Dolby Media Decoder** > **Preferences**.
2. In the **Preferences** window, click the **Speaker Settings** tab.
   The **Speaker Settings** page displays.

Related information

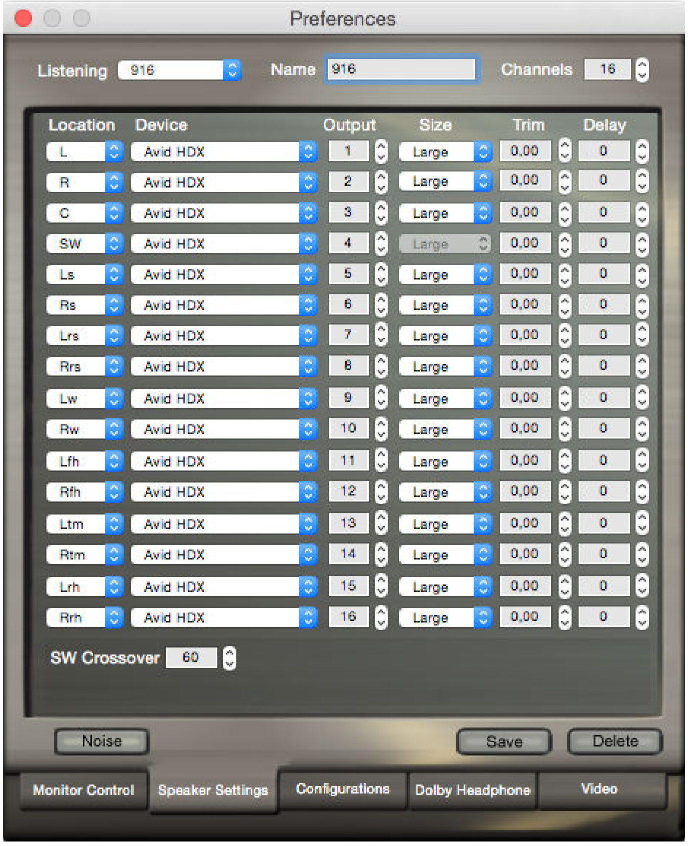
Setting up Dolby Media Decoder on page 10

Monitor panel on page 29

3.3 Creating listening modes

You can create a custom listening mode that will be available in the **LISTENING** section of the monitor panel.

Figure 2: Example setting for a listening mode



Procedure

1. Open the **Speaker Settings** page.
2. From the **Listening** drop-down menu, select an unused slot.
3. In the **Channels** field, select the number of output channels required (2 – 16).

   Note: L and R are required. A subwoofer (SW) output must be added if defining additional channels besides L and R.

4. Select a **Location**, **Device**, **Output**, **Size**, **Trim**, and **Delay** value for each listed channel.
5. For modes using a subwoofer, select a **SW Crossover** value.
6. In the **Name** field, enter a name for the mode.

   This name appears as the button name in the **LISTENING** section of the monitor panel after you save the listening mode.

7. Click **Save**.
   The configuration is saved and is available as a new listening mode button in the **LISTENING** section of the monitor panel.

Related information

Listening mode parameters on page 12

3.3.1 Listening mode parameters

When creating a new listening mode, you have to select listening mode parameters such as **Location**, **Device**, **Output**, **Size**, **Trim**, **Delay**, and **SW Crossover** value.

Location

Selecting an output channel from the **Location** drop-down menu assigns a specific rendered audio signal to a specific I/O output, such as Left (L) to output channel 1, and Right (R) to output channel 2.

Device

Use to select the I/O device to be used for each channel. The decoder defaults to the built-in output of your computer. Only properly set-up devices recognized by the system are listed.

Output

Use to select the output number of the selected I/O device to be used by the channel. Two channel IDs cannot be assigned to the same output channel.

Size

The **Size** setting determines whether to use bass management for the channel.

Large indicates that the speaker is considered full range and does not require bass management.

Small indicates that the speaker is a near-field or satellite design and is not designed to reproduce extremely low frequencies (typically, frequencies below 80 Hz). When a channel is set to **Small**, the system filters out low-frequency information from that channel and redirects it to the subwoofer output. If no subwoofer is present in a listening mode, the selection of **Small** (or no bass management) is not allowed, and the **Size** field is unavailable. The subwoofer is required for layout speakers greater than two-speakers.

Trim

Use this control to trim output channels during sound system calibration. The range is quarter-decibel steps from 0 to –30 dB.

Delay

This setting controls the amount of delay (in milliseconds) that is added to the output channel for use in time aligning your sound system.

Tip: Assign the furthest channel 0 delay, and add 1 ms delay to each channel for each foot closer they are to the reference position.

Subwoofer Crossover

The appropriate setting for the subwoofer crossover frequency depends on the capability of your front and surround speakers. When any channels are set to **Small** on the **Speaker Settings** page, the decoder diverts signals below the subwoofer crossover value to the subwoofer output. Most smaller speakers require an 80 Hz subwoofer crossover, but you should use the setting most appropriate for your speakers. In a system with all speakers set to **Large**, the crossover value is not applied.

Related information

Creating listening modes on page 11

Output channel locations on page 13

Internal and external audio hardware on page 17

3.3.2 Output channel locations

Selecting an output channel from the Location drop-down menu assigns a specific rendered audio signal to a specific I/O output, such as Left (L) to output channel 1, and Right (R) to output channel 2.

Refer to this table for standard labels for the supported speaker channel locations and their descriptions.

Table 2: Channel location labels and descriptions

Location label	Description
L	Left, a loudspeaker position behind the screen to the far-left edge.
R	Right, a loudspeaker position behind the screen to the far-right edge.
C	Center, a loudspeaker position behind the screen, corresponding to the horizontal center of the screen.
SW	Subwoofer, a band-limited low-frequency-only loudspeaker positioned at the screen end of the room. Delivers LFE content, plus any redirected bass from bass-managed (small) speakers.
Ls	Left Surround, a loudspeaker (or an array in a 5.1 or 7.1 cinema system) along the left side of the room.
Rs	Right Surround, a loudspeaker (or an array in a 5.1 or 7.1 cinema system) along the right side of the room.
Lrs	Left Rear Surround, a loudspeaker (or an array in a 5.1 or 7.1 cinema system) on the back wall of the room in the left corner.
Rrs	Right Rear Surround, a loudspeaker (or an array in a 5.1 or 7.1 cinema system) on the back wall of the room in the right corner.
Cs	Center Surround, a loudspeaker centered on the back wall of the room.
Lc	Left Center, a loudspeaker position midway between the center of the screen and the left edge of the screen.
Rc	Right Center, a loudspeaker position midway between the center of the screen and the right edge of the screen.
Lw	Left Wide, a loudspeaker position outside the screen area, at the far-left front in the room.
Rw	Right Wide, a loudspeaker position outside the screen area, at the far-right front in the room.
Lfh	Left Front Height, an overhead loudspeaker for Dolby Atmos, at the left front of the room; Lvh is treated as Lfh.
Rfh	Right Front Height, an overhead loudspeaker for Dolby Atmos, at the right front of the room; Rvh is treated as Rfh.
Ltf	Left Top Front, an overhead loudspeaker for Dolby Atmos, at the left front of the room.
Rtf	Right Top Front, an overhead loudspeaker for Dolby Atmos, at the right front of the room.
Ltm	Left Top Mid, an overhead loudspeaker for Dolby Atmos, at the left middle of the room.
Rtm	Right Top Mid, an overhead loudspeaker for Dolby Atmos, at the right middle of the room.
Ltr	Left Top Rear, an overhead loudspeaker for Dolby Atmos, at the left rear of the room.
Rtr	Right Top Rear, an overhead loudspeaker for Dolby Atmos, at the right rear of the room.
Lrh	Left Rear Height; Dolby Digital Plus bitstreams carrying Dolby Atmos audio and associated metadata also support Lfh/Rfh (or Lvh/Rvh) left/right front height speaker locations. These locations are also supported by Dolby Digital, Dolby Digital Plus, and Dolby True HD bitstreams.
Rrh	Rrh Right Rear Height; Dolby Digital Plus bitstreams carrying Dolby Atmos audio and associated metadata also support Lfh/Rfh (or Lvh/Rvh) left/right front height speaker locations. These locations are also supported by Dolby Digital, Dolby Digital Plus, and Dolby True HD bitstreams.
Lsc	Left Surround Center, a loudspeaker along the middle of left side of the room.

Table 2: Channel location labels and descriptions (continued)

Location label	Description
Rsc	Right Surround Center, a loudspeaker along the middle of the right side of the room.
Ls1	Left Surround 1, a loudspeaker along the left side of the room.
Rs1	Right Surround 1, a loudspeaker along the right side of the room.
Ls2	Left Surround 2, a loudspeaker along the left side of the room.
Rs2	Right Surround 2, a loudspeaker along the right side of the room.
Lrs1	Left Rear Surround 1, a loudspeaker on the back wall of the room in the left corner.
Rrs1	Right Rear Surround 2, a loudspeaker on the back wall of the room in the right corner.
Lrs2	Left Rear Surround 2, a loudspeaker on the back wall of the room in the left corner.
Rrs2	Right Rear Surround 2, a loudspeaker on the back wall of the room in the right corner.
Lcs	Left Center Surround, a loudspeaker along the center of the left side of the room.
Rcs	Right Center Surround, a loudspeaker along the center of the right side of the room.

The following illustrations provide relative positioning of floor and overhead speakers. Distances and angles are not to scale. To find detailed speaker layout guidance for both traditional and Dolby Atmos home theaters, refer to the information provided on the Dolby website: http://www.dolby.com/us/en/guide/speaker-setup/index.html.

Figure 3: Relative positions of floor speakers (not to scale)

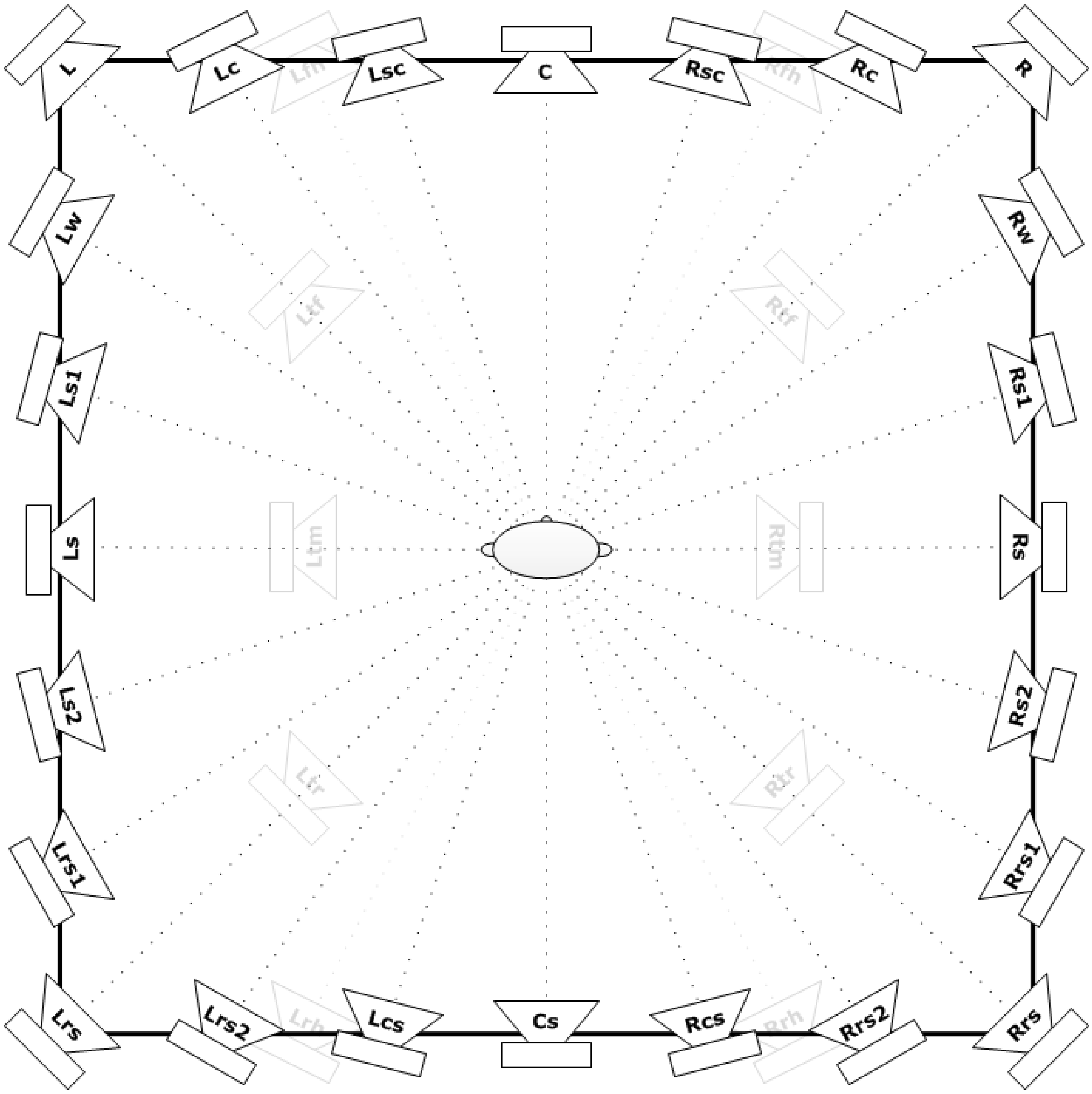
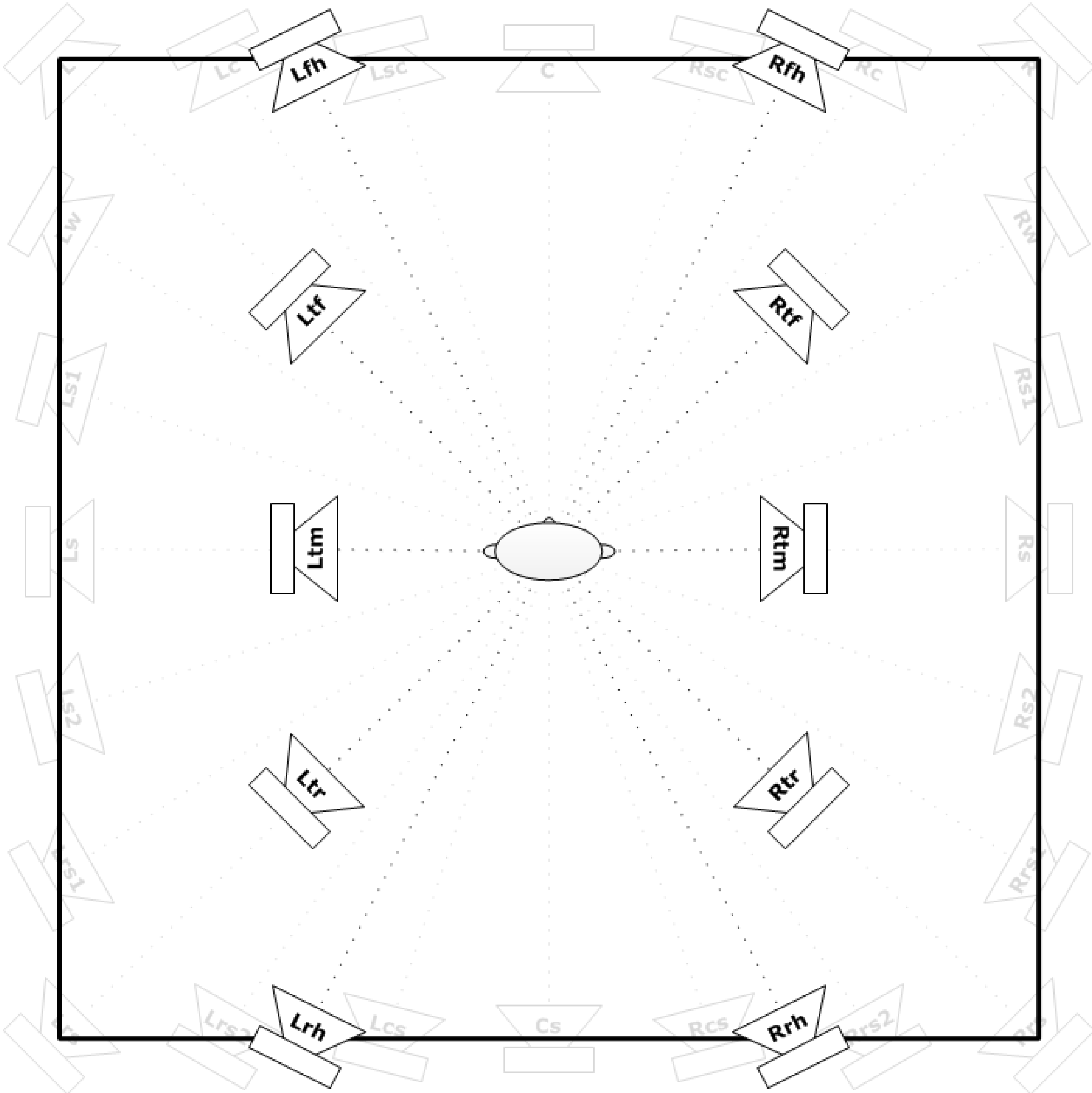


Figure 4: Relative positions of overhead speakers (not to scale)



Related information

Listening mode parameters on page 12

3.3.3 Internal and external audio hardware

Dolby Media Decoder automatically detects built-in audio hardware and most audio devices attached to the computer. These devices include Thunderbolt, FireWire, and USB interfaces. Occasionally, additional setup is necessary before the Dolby Media Decoder recognizes an I/O device.

Consider the following provisions when working with external audio hardware:

- Make sure that the I/O driver (if required) is properly installed. For the latest drivers, refer to the manufacturer's website.
- If the driver is properly installed, but is not recognized, reboot your computer.
- Refer to your I/O documentation, as needed. Always make sure you are using the latest firmware and driver (if needed) for your audio hardware.

- Many audio hardware devices provide a number of settings and controls, including hardware setup options and output volume control, which can affect audio playback from Dolby Media Decoder. Check these settings carefully.
- Make sure the audio device is powered on before launching Dolby Media Decoder.
- If using an Avid I/O as your audio hardware, make sure that Pro Tools is not currently using the Avid I/O before attempting to access it from the Dolby Media Decoder player.

Note: The decoder cannot access an I/O device that is in use by any other program. Make sure that neither Avid Pro Tools nor any other program is currently running and accessing the I/O.

The setup programs for Avid and other interfaces can alter the analog, AES, and S/PDIF output assignments. Make sure the channel mapping on your Pro Tools or other I/O device is correct.

Related information

Setting up Dolby Media Decoder on page 10

Listening mode parameters on page 12

3.3.4 Testing a listening mode

Dolby Media Decoder provides easy ways of testing listening modes (for example, to perform required speaker calibration).

Procedure

1. On the **Speaker Settings** page, click **Noise**.

   The system automatically cycles the noise through all of the channels present within the current listening mode.
2. If a channel needs level adjustment, click its **Trim** dial.

   The system automatically locks the noise to the channel being trimmed.
3. Adjust other channels as required.
4. When you are satisfied with all of the channels, click **Noise** to turn noise off.

3.3.5 Changing monitor control settings

The options on the **Monitor Control** page of the **Preferences** window determine the behavior of the controls in the monitor panel. For example, the behavior of the **Custom** controls in the **Custom Compression** section of the monitor panel is controlled here, along with several other aspects of the decoder behavior.

About this task

The options on this page are the same as the options available in most consumer receivers. Use the options to emulate a broad range of consumer experiences in the studio.

Procedure

1. Select **Dolby Media Decoder** > **Preferences**.
2. Click the **Monitor Control** tab.
3. Configure the **Pro Logic II** settings as needed.
4. Configure the **Decode Control** settings as needed.
5. Configure the **Custom Compression** settings as needed.

   Settings are remembered when you move to another page.

Related information

Pro Logic II settings on page 19
Decode Control settings on page 20
Custom Compression settings on page 20

Pro Logic II settings

Dolby Pro Logic II technology processes any high-quality stereo (two-channel) audio into five playback channels of full-bandwidth surround sound.

To begin configuring Dolby Pro Logic II decoding, select an option in the **Mode** field. The options are described in the following table.

Table 3: Dolby Pro Logic II modes

Mode	Description
Movie	The default mode. The **Dimension**, **Panorama**, and **Center Width** values are not applied to the audio when the **Movie** mode is selected.
Music	Selection of **Panorama** is allowed. Changing the **Dimension** and **Center Width** values is also allowed in **Music** mode.
Matrix	Similar to the **Movie** mode. When **Matrix** is selected, the decoder emulates a traditional, simple passive matrix decoder. Some very old legacy content was designed for passive matrix decoders, which exhibit a very limited level of separation between adjacent output channels of about 3 dB.

Note: Depending on the mode that is selected, a Pro Logic delay is added to the surround delay value. See Table 6 on page 31 for more details.

The following three controls apply only when the **Music** mode is used. The **Movie** mode is designed to provide a very specific consistent experience for films. It gives the music listener more control over the surround effect because the source music may be created with a very wide range of ambience and stereo widths.

Panorama

This option brings additional ambience and other sound elements from the front sound stage into the rear channels for a more immersive surround experience from music content. It can be either enabled or disabled, depending on the listener's preference.

Note: The type and location of the side speakers often have a large impact on the effect of this parameter.

Center Width

This option adjusts the spread of the Center channel from hard center to full left and right, with no signal sent to the center speaker. With this control, you can create a balanced left-center-right stage presentation for both the driver and the front passenger in a vehicle. For home users, it allows improved blending of the center and main speakers and provides control the sense of image width (or weight). The decoder provides single-step values from −3 to +3. The default value is 0.

Dimension
: This option adjusts the front-to-back balance of the surround channels. This can help achieve a more suitable balance from all of the speakers with certain recordings. The decoder provides single-step values from −3 to +3. The default value is 0.

Related information

Changing monitor control settings on page 18

UPMIXING section on page 31

Decode Control settings

The **Dolby Control** settings determine whether the decoding type is set automatically or manually.

*Table 4: **Decode Control** options*

Control	Options and effects	
	Auto	**Manual**
Pro Logic Decode	Causes the system to determine whether to engage Dolby Pro Logic or Pro Logic II decoding, based on the state of the two-channel format (Surround mode) flag, set within the metadata of a two-channel encoded file.	Requires the user to manually enable Dolby Pro Logic or Pro Logic II decoding within the **UPMIXING** section.
Dolby Digital EX Decode	Causes the system to determine whether to engage Dolby Digital Surround EX decoding based on the state of the Dolby Digital Surround EX flag set within the metadata of an encoded file that contains both Ls and Rs channels.	Requires the user to manually enable Dolby Pro Logic or Pro Logic II decoding within the **UPMIXING** section.

Related information

Changing monitor control settings on page 18

Custom Compression settings

The **Custom Compression** settings (**Boost** and **Cut**) determine the levels of boost and cut that are applied when you click **Custom** in the **Compression** section of the monitor panel.

This feature emulates a consumer receiver that allows customization of Line mode compression. Boost and cut can both be adjusted in tenths from 0.0 to 1.0. The default value for each is 1.0.

Related information

Changing monitor control settings on page 18

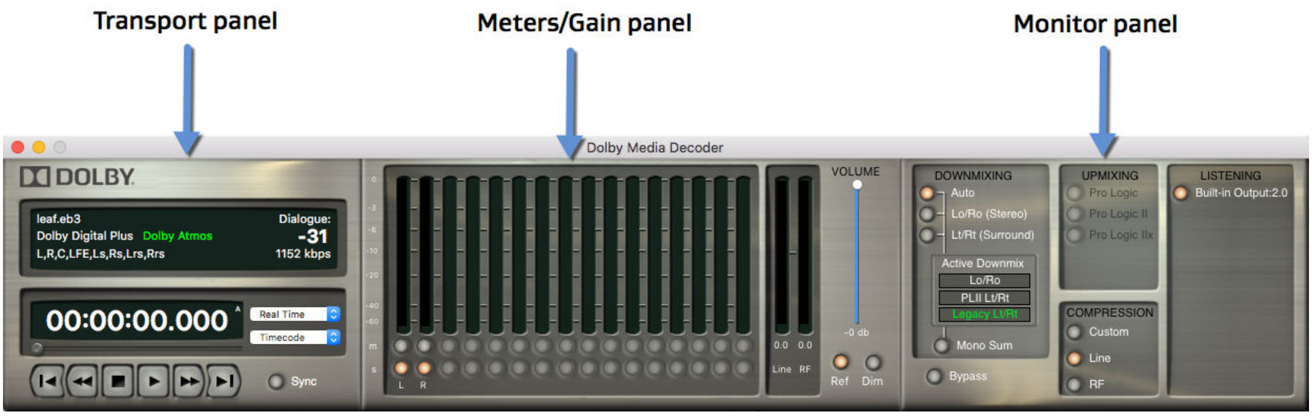
COMPRESSION section on page 32

4 Dolby Media Decoder operation

With Dolby Media Decoder, you can load and play back audio files, synchronize them with a video content, and monitor playback levels.

- Loading audio files for playback
- Setting up video for synchronous playback in QuickTime Movie
- Saving configuration files

Figure 5: Dolby Media Decoder interface



4.1 Loading audio files for playback

You can load and play back an encoded (multichannel interleaved) file with Dolby Media Decoder.

About this task

When loading Dolby TrueHD bitstreams, visual feedback of loading activity displays in the **Building Index for Audio File** window. You can click **Cancel** in this window to stop loading. When a file is loaded, the file name and other pertinent information are displayed in the information display in the transport panel.

Procedure

To load an encoded file, do one of the following actions:

- Drag and drop an audio file onto the information display in the transport panel.
- Double-click the information display in the transport panel. In the **Open** window, browse to the location of the file, highlight it, and click **Open**.

- Select **File** > **Open File**. In the **Open** window, browse to the location of the file, highlight it, and click **Open**.
- If that extension has Dolby Media Decoder set as the default application in the Finder, double-click an encoded file in the Finder window.

4.2 Setting up video for synchronous playback in QuickTime Movie

The decoder supports synchronizing video to audio via a QuickTime video file. Any video format supported by QuickTime may be used as long as it is in a .mov container.

About this task

When audio and video files begin or end at different times, the decoder starts playback with the file that starts first and stops playback at the end of the last file to ensure that the full timeline is decoded.

The QuickTime movie screen includes a playback media indicator in the footer bar: **Audio**, **Video**, or **Audio/Video**.

Procedure

1. From the menu bar, load the QuickTime movie by doing one of the following actions:
   - Select **File** > **Open Movie**, and then browse to the movie you want to open.
   - Drag and drop a QuickTime movie into the information display in the transport panel.
   - Drag and drop a QuickTime movie into the QuickTime movie screen window.

   The system opens the selected file in the QuickTime movie screen window.

   Tip: If your QuickTime movie is an anamorphic video, you can disable the **Lock Aspect Ratio** button to unlock it. Unlocking the aspect ratio allows you to manually stretch the image for improved viewing.
2. Click the **TC Start** button in the lower-left corner of the movie screen window. The system opens the **QuickTime Timecode Start** dialog box.
3. In the timecode format drop-down menu, select the correct format.
4. In the **Timecode Start** field, enter the required timecode start (first frame of the video).
5. To accept the settings, click **OK**.

   Tip: You must supply the correct frame rate and the correct value for the first frame of timecode for the QuickTime movie to allow synchronous playback with audio.
6. When the setup is complete, click **Sync** in the transport panel to activate synchronization.

   Note: The **Sync** button is enabled by default when you load audio and video.

Related information

Transport panel on page 25
Sync button on page 27

4.3 Saving configuration files

Saving decoder configuration files, such as snapshots of the current state of the pages in the Preferences window, can eliminate repetitive selection of multiple settings. This can be particularly helpful when multiple users have their own preferred speaker settings.

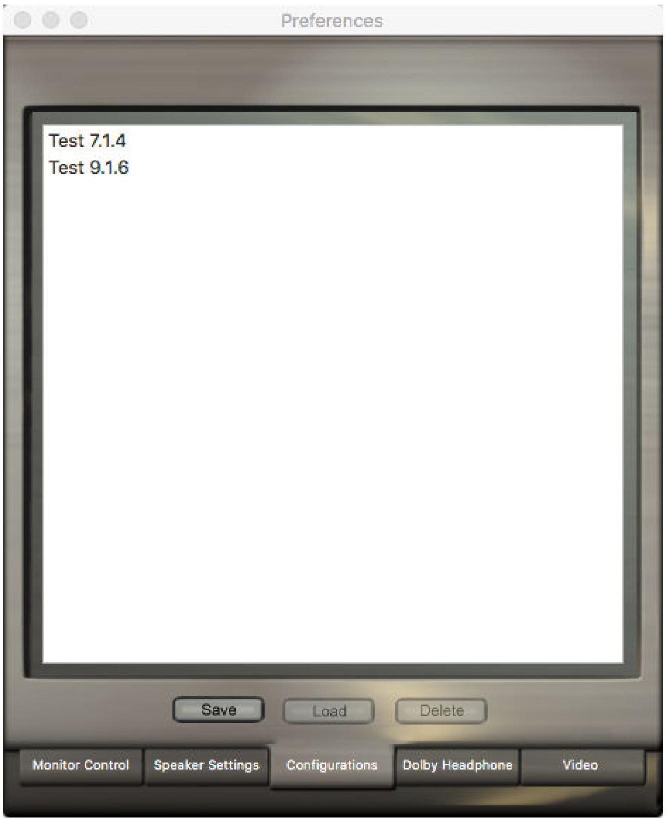
About this task

Saved decoder configuration files can be loaded with one click. There is no limit on the number of decoder configuration files you can save.

Procedure

1. Select **Dolby Media Decoder** > **Preferences**.
2. Click the **Configurations** tab.
   The **Preferences** page listing all previously saved decoder configuration files appears.

   Figure 6: Preferences page

3. Click **Save**.
4. Enter a name for the decoder configuration file. Then browse to a storage location, and click **Save**.

   Notice: Configuration files appear on the list only if they were saved in a default location (Library/Application Support/Dolby). We recommend not changing the default storage location.

Note: We recommend saving the default state of the application as the first decoder configuration file.

We also recommend using clear, descriptive names for decoder configuration files to minimize search time.

What to do next
You can load or delete configuration files using buttons below the list.

Related information

Setting up Dolby Media Decoder on page 10

5 Application user interface

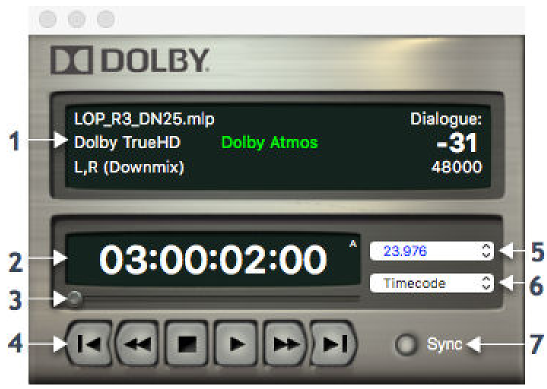
The Dolby Media Decoder user interface consists of three panels that provide control and monitor functions.

- Transport panel
- Meters/gain panel
- Monitor panel
- Metadata window

5.1 Transport panel

The transport panel displays information about the loaded material and provides controls for playback.

Figure 7: Transport panel



The transport panel includes the following displays and controls:

1. Information display

Provides file name and other relevant information, depending on the type of file loaded.

The display includes:

File name
Displays the name of the file (including its extension) on the first line.

Bitstream type

Displays the Dolby audio format (Dolby TrueHD, Dolby Digital Plus, Dolby Digital, or MLP) on the second line. For bitstreams carrying Dolby Atmos audio and metadata, the bitstream type is followed by a Dolby Atmos indication.

The Dolby Atmos indication text is green for all bitstreams with Dolby Atmos objects and metadata. The indication is orange if the additional `ADDBSI` header information in Dolby Digital Plus is not present. The decoder supports playback for both signals.

(Downmix) indication
Confirms Dolby TrueHD or MLP presentations that are downmixes. This indication is on the third line.

Dialogue
Provides the dialogue normalization parameter on the right side of the information display.

Data-rate indication
Displays the current data rate (in kbps) on the right side of the information display, below the dialogue normalization parameter.

2. Timecode counter

Displays timecode, based on the current display type and mode. The counter is optimized to be audio sample accurate when entering a time value.

When the display mode is set to **Timecode** or **File Position**, the counter displays the current timecode. When the display mode is set to **File Start** or **File End**, the counter is fixed to the value for the set mode.

The timecode counter includes a playback media indicator (upper-right corner of the timecode counter): A for audio only, V for video only, or AV for audio/video.

3. Timecode slider
Lets you change the position of playback material. The slider is optimized to be audio sample accurate. When moving the slider, the following changes occur:

- The timecode counter shows the current position.
- Video updates to the current position.
- The meter display clears, and there is no sound.

4. Transport controls
Provides controls for playback of audio, QuickTime movies, and connected external devices. Controls include: first frame, rewind, stop, play, fast forward, and last frame.

5. Display type

Lets you set how the file position is displayed for both audio and QuickTime files.

The supported display types are:

- Timecode: 23.976, 24, 25, 29.97, and 29.97 df, and 30 fps
- Real time
- Seconds

When an audio file is opened, the decoder automatically sets the display type to the frame rate of the timecode embedded within the file and displays the starting timecode. If no timecode is embedded in the file, the decoder switches to the real time value (in hour:minutes:seconds.milliseconds).

6. Counter mode
Lets you decide how your audio and video are aligned on the timeline and which type of information is displayed in the counter. The supported counter modes are:

Timecode
Displays timecode embedded in the audio or QuickTime movie file. If both audio and video files are loaded, it aligns them on the timeline based on the embedded timecodes. With **Timecode** as the counter mode, the first timecode frame is the value embedded in the first frame of the file, which might or might not be 00:00:00:00.

File Position
Displays file position according to the selected display type with a zero reference start value. With **File Position** as the counter mode, the first timecode frame is always 00:00:00:00.

File Start
Displays the timecode embedded in the first frame of the file.

File End
Displays the timecode embedded in the last frame of the file.

7. Sync button
Lets you set which devices the transport panel controls.

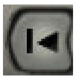
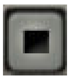
Related information

Setting up video for synchronous playback in QuickTime Movie on page 22
Transport controls on page 27
Sync button on page 27
Sync button on page 27

5.1.1 Transport controls

You can use the transport controls in the transport panel to control the playback of Encoded Dolby audio files or QuickTime–wrapped video files (.mov).

Table 5: Decoder transport controls

Button	Name	Function
	Home (first frame)	Positions the decoder at the first frame of the open file. Keyboard shortcut: Command+Left Arrow.
	Rewind	Moves backward in time through the open file. Each time you click the button, the movement accelerates. Keyboard shortcut: Left Arrow.
	Stop	Ends playback and returns the decoder to the last position set by the timecode slider or to the first frame of the open file if the timecode slider was not used. Keyboard shortcut: Enter (toggles Play/Stop).
	Play/pause	Keyboard shortcut: Enter (toggles Play/Stop), or space bar (toggles Play/Pause).
	Fast forward	Moves forward in time through the open file. Each time you click the button, movement accelerates. Keyboard shortcut: Right Arrow.
	End (last frame)	Positions the decoder at the last frame of the open file. Keyboard shortcut: Command+Right Arrow.

Related information

Transport panel on page 25

5.1.2 Sync button

The state of the **Sync** button determines which devices the decoder controls.

The following control states are indicated by the **Sync** button:

- When the button is not lit, the transport controls the playback of loaded audio files only.
- When the **Sync** button is lit, the transport controls both audio playback from Dolby Media Decoder and one video source simultaneously.

Click (highlight) a button to enable it. Click again to remove highlighting it and disable it.

The **Sync** button can then be switched off, if desired, to allow the transport to control the video only.

Sync is enabled automatically when both an audio and a .mov files are loaded.

Note: Synchronous playback requires correctly specified timecode and frame rates for both audio and video.

When audio and video files begin or end at different times, the decoder starts playback at the earliest file and stops playback at the end of the last file to ensure that the full timeline is decoded.

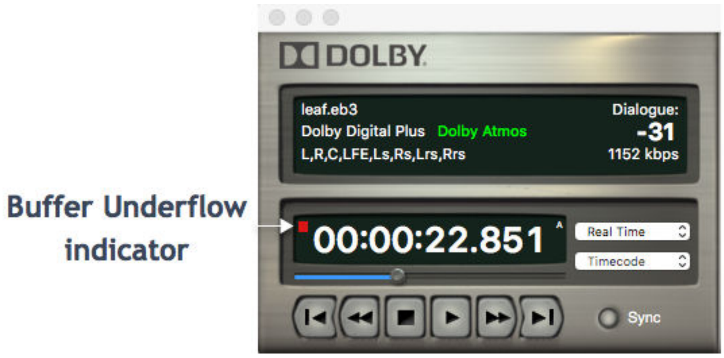
Related information

Transport panel on page 25
Transport panel on page 25
Setting up video for synchronous playback in QuickTime Movie on page 22

5.1.3 Buffer underflow indicator

The buffer underflow indicator turns red when Dolby Media Decoder is unable to get the input data from the loaded files quickly enough, or is unable to decode it fast enough to avoid gaps in the audio output.

Figure 8: Buffer underflow indicator in the transport panel

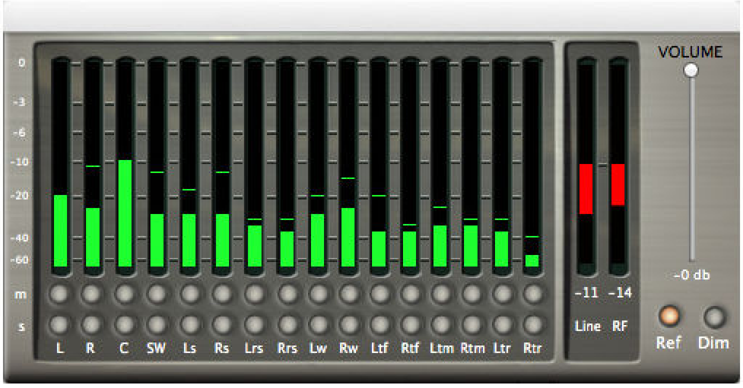


If this indicator is visible, we recommend freeing up system memory and CPU resources by closing other open applications, widgets, and any background processes that require CPU time or memory. In some cases, a faster hard-disk drive, faster network connection (for remote servers or remote file directories), or more computer RAM may be needed to ensure smooth, uninterrupted audio playback of complex multichannel content.

5.2 Meters/gain panel

The meters/gain panel displays playback levels as well as the current channel configuration. Additionally, the panel includes controls for managing the volume and mute or solo each channel.

Figure 9: Meters/Gain panel



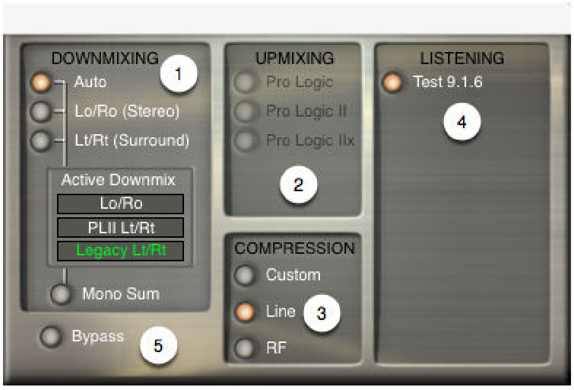
The meters/gain panel provides:

- 16-channel metering that supports speaker configurations of up to 16 speakers. Metering labels are at 0, –3, –6, –10, –20, –40, and –60 dB levels. The signal is displayed as follows:
    - In green when below –6 dB
    - In yellow from –6 to –0.000005 dB
    - In red when at or above 0 dB
- **Line** and **RF** meters that display the cut or boost to be apply if downmixing is enabled or if a late-night listening mode is selected (Line or RF mode). The scale for the RF boost and cut meter bar is compressed because RF mode compression can vary over a wider range of values. When playing Dolby TrueHD files, the Line mode and custom mode compression are available, but the RF mode compression is not.
- **m** button that lets you mute each channel and **s** button that lets you solo each channel.
- **Volume** slider to adjust the attenuation. The slider includes a digital display showing the attenuation value (in dBFS).
- **Ref** button that stays lit as long as the master volume is 0 dB. If the slider is moved, the button becomes disabled. If you click the **Ref** button, the master volume is reset to 0 dB.
- **Dim** button that applies 20 dB of attenuation when enabled.

5.3 Monitor panel

The monitor panel provides controls for playing back downmixes, applying compression, and selecting which user-created listening mode (speaker settings) to use.

Figure 10: The monitor panel



The Monitor panel consists of following elements:

1. **DOWNMIXING** section, where you can choose to listen to a downmix of an overall mix.
2. **UPMIXING** section, which lets you select and listen to upmixes of the loaded multichannel audio.
3. **COMPRESSION** section with compression profiles.
4. **LISTENING** section with saved speaker configurations.
5. **Bypass** button to bypass all downmix, upmix, and compression processing.

Related information

Setting speaker configurations on page 10

5.3.1 DOWNMIXING section

When multichannel Dolby Digital and Dolby Digital Plus encoded files are loaded, you can listen to downmixes of the overall mix. The available choices are determined by the format.

The following downmixing modes are available with Dolby Digital and Dolby Digital Plus:

- **Auto**
- **Lo/Ro (Stereo)**
- **Lt/Rt (Surround)**
- **Mono Sum**

Mono can be enabled for a two-channel downmix only when two-channel downmixing is active. We recommend enabling the Lo/Ro downmix before enabling Mono Sum mode. Use of an Lt/Rt downmix results in the loss of surround content during a mono summation.

Active downmix

The **Active Downmix** section displays the actual downmix used for the current file. The specific type of downmixing applied is determined by a combination of user settings, bitstream type, and bitstream metadata. In Auto mode, a flag in the bitstream may indicate that an Lo/Ro or an Lt/Rt downmix is preferred by the content creator. Dolby Digital Plus bitstreams, which are encoded with proper downmix overload protection for a Dolby Pro Logic II compatible downmix, have Dolby Pro Logic II Lt/Rt downmixing enabled when the user selects Lt/Rt mode. Dolby Digital Plus bitstreams that have a metadata flag set for legacy Lt/Rt as the preferred downmix, and all Dolby Digital bitstreams will have the legacy Lt/Rt downmix enabled when the user selects Lt/Rt mode.

Downmixes with Dolby TrueHD

With Dolby TrueHD, downmixing is handled by the encoder only. This differs from Dolby Digital and Dolby Digital Plus, where downmixing is handled by the decoder.

Dolby TrueHD playback allows to listen to individual presentations. Therefore, when decoding Dolby TrueHD, the **DOWNMIX** section is replaced by a **PRESENTATIONS** section.

Figure 11: Monitor panel with the PRESENTATIONS section

This allows you to listen to either the two-, six-, or eight-channel presentation within a Dolby TrueHD stream. The **Presentations** section is set to the largest independent presentation as the default option within the selected Dolby TrueHD bitstream, as follows.

- Dolby Atmos presentation is selected upon loading a Dolby TrueHD with Dolby Atmos file if the listening configuration is greater than 5.1 speaker channels.
- Eight-channel presentation is selected upon loading a channel-based Dolby TrueHD file if the listening configuration is greater than 5.1 speaker channels, up to 7.1 speaker channels.
- Six-channel presentation is selected upon loading any Dolby TrueHD file (with or without Dolby Atmos) if the listening configuration is greater than two speaker channels, up to six (5.1) speaker channels.
- Two-channel presentation is selected upon loading any Dolby TrueHD file (with or without Dolby Atmos) if the listening configuration has two speaker channels.

5.3.2 UPMIXING section

You can listen to upmixes of the loaded multichannel audio, including an upmix of a downmix.

Upmixing of encoded audio files can be auditioned by clicking on any of the buttons in the **UPMIXING** section of the Monitor panel.

The following modes are available in the **UPMIXING** section:

- Dolby Surround / Dolby Pro Logic Technology
- Dolby Pro Logic II technology
- Dolby Pro Logic IIx technology

When one of the upmixing modes is used, a fixed Pro Logic delay value is automatically added to the surround delay value. The value depends on the upmixing mode and the Dolby Pro Logic II mode (only for Dolby Pro Logic II and IIx). The following table shows delay offsets (in ms) that are applied to audio signals.

Table 6: Delay offsets

Upmixing mode	Dolby Pro Logic II mode	C signal delay [ms]	Ls/Rs signals delay [ms]	Lb/Rb or Cs signals delay [ms]
Dolby Pro Logic	-	0	10	-
Dolby Pro Logic II	Movie	0	10	-
Dolby Pro Logic IIx		0	10	20

Table 6: Delay offsets (continued)

Upmixing mode	Dolby Pro Logic II mode	C signal delay [ms]	Ls/Rs signals delay [ms]	Lb/Rb or Cs signals delay [ms]
Dolby Pro Logic II	Music	2	0	-
Dolby Pro Logic IIx		2	0	10

Related information

Dolby Surround / Dolby Pro Logic technology on page 43
Dolby Pro Logic II technology on page 43
Dolby Pro Logic IIx technology on page 44
Pro Logic II settings on page 19

5.3.3 COMPRESSION section

The **COMPRESSION** section includes compression profiles, which can be applied to the Dolby TrueHD, Dolby Digital Plus, or Dolby Digital bitstream during decoding playback. These compression presets aid playback for two-channel downmix overload protection and late-night listening modes.

Compression profiles include:

Custom mode
: Emulates a consumer receiver that allows customization of Line mode compression during decoding playback. Boost and cut can both be adjusted in tenths from 0.0 to 1.0. The default value is 1.0. For information on creating a custom compression profile on the **Monitor Control** page of the **Preferences** window, see Changing monitor control settings on page 18.

Line mode
: Applies light compression characteristics during decoding playback. See sections below for more details.

RF mode
: Applies heavy compression characteristics during decoding playback. See sections below for more details.

When playing Dolby TrueHD files, only Line mode and custom mode compression are available.

Line mode

In Line mode, the normalized dialogue level is reproduced from the decoder at a constant loudness level of –31 dBFS Leq(A), assuming the dialogue level parameter is set correctly.

Line mode compression is designed to provide gentle, musical compression appropriate for the dynamic range expected from consumer line-level or power-amplified outputs from two-channel set-top decoders, two-channel digital televisions, 5.1-channel digital televisions, and Dolby Digital audio/video surround decoders.

Consumer control of the dynamic range is limited when downmixing. Products with stereo or mono outputs do not usually allow consumer scaling of Line mode. This is because these devices are usually downmixing (for example, when receiving a 5.1-channel signal).

However, in these products, the consumer may have a choice between Line mode and RF mode.

RF mode

In RF mode, high- and low-level compression scaling is not allowed. When RF mode is active, that compression profile is always fully applied.

RF mode was originally designed for products (such as set-top boxes) that generated a downmixed signal for connection to the RF/antenna input of a television set; however, it is also useful in situations where heavy dynamic range compression is required (for example, when small PC speakers are used for DVD playback, for late-night listening, or anytime the user is in an environment with limited useful dynamic range or high noise levels). In RF mode, the overall program level is raised 11 dB. This results in dialogue being reproduced at an average level of –20 dBFS Leq(A), while the peaks are limited to prevent signal overload in the D/A converter. By limiting headroom, severe overmodulation of television receivers is prevented. The 11 dB gain provides an average loudness level that compares well with existing analog television broadcasts and is therefore used to match the levels of decoded TV content to other programs on a broadcast service.

For some types of speech-heavy content, it may be necessary to further constrain signal peaks above the average dialogue level so that there is less than 20 dB headroom. The selection of a suitable RF mode profile should be used during encoding.

Related information

Custom Compression settings on page 20
Setting up Dolby Media Decoder on page 10

5.3.4 Bypass button

You can bypass all processing for downmixing and dynamic range (except for dialogue normalization scaling) by clicking **Bypass** in the monitor window.

When the decoder is in **Bypass** mode, none of the selected downmixing or compression is audible, and these sections are unavailable. Dialogue normalization is still applied. Settings on the **Speaker Settings** page are still active.

5.4 Metadata window

The metadata window displays file information and metadata for the current open file. Each type of encoded file contains a different set of metadata parameters, so this window looks different depending on which file type is loaded.

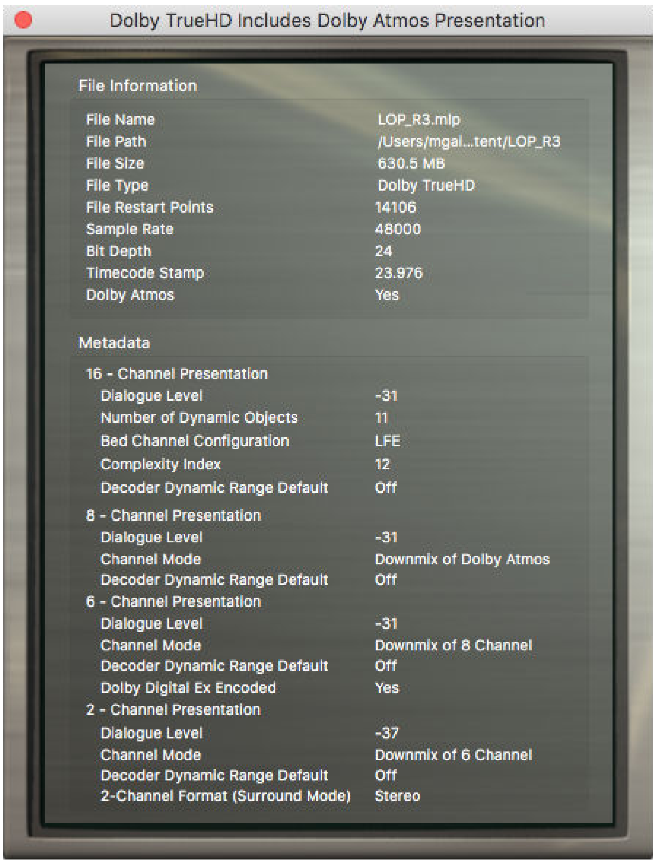
To open the metadata window (or close when open), do one of the following actions:

- From the menu bar, select **Window** > **Metadata**.
- With the decoder menu in focus, press M on the keyboard.

Figure 12: Metadata window for Dolby Digital Plus with Dolby Atmos files

The metadata window headers display the type of encoded file, for example, “Dolby Digital Plus with Dolby Atmos” for Dolby Digital Plus with Dolby Atmos encoded files.

Figure 13: Metadata window for Dolby TrueHD files including Dolby Atmos presentations



The Dolby Atmos field indicates whether the encoded file bitstream includes Dolby Atmos content.

For Dolby TrueHD including a Dolby Atmos presentation or Dolby Digital Plus with Dolby Atmos encoded files, the metadata section additionally presents the Number of Dynamic Objects, information on Bed Channel Configuration, and a Complexity Index presenting the total sum of bed channels and dynamic objects present in the bitstream.

6 Speaker system calibration

Before decoding audio, you must calibrate the monitoring system to establish a balance between all channels and to ensure that all speakers play back at the correct level relative to the listening reference position.

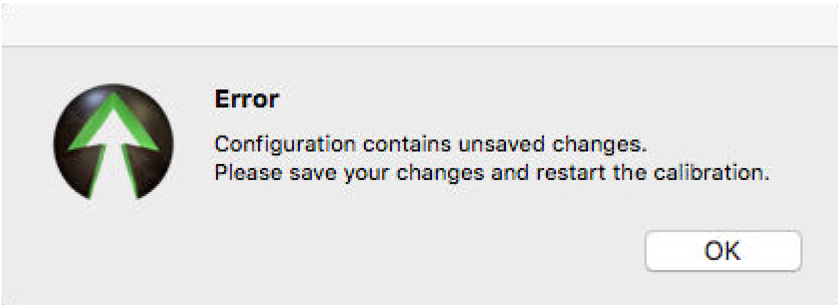
6.1 Speaker calibration functionality

The built-in speaker calibration provides specific functionality as you enable noise, calibrate noise, and modify trims and delay controls during calibration.

Consider the following points before you enable noise for speaker calibration:

- Initial speaker calibration:
    - If the **Speaker Settings** page contains unsaved changes, you will not be able to start calibration. A following error appears.

      Figure 14: Speaker calibration error

    - Clicking **Noise** starts the current listening layout displayed at the top of the **Speaker Settings** page in the **Preferences** window.

      If the selected listening mode is broken or not set, an error message displays and the noise generation stops.
    - Calibration always initially starts in an automatic mode. In this mode, noise continuously switches through each speaker, in order of their presence in the meter display.
- Decoder functionality during noise calibration:
    - Transport controls as well as **Downmixing**, **Upmixing**, **Bypass**, **Compression**, and presentation controls are disabled.
    - Opening or closing a file is not allowed.
    - Changing listening mode on the **LISTENING** page is disabled.
    - Switching the listening mode layout does not save changes made to **Trim** or **Delay**. You must manually save the changes by clicking **Save**.
    - Clicking **Save** stops calibration and saves the listening layout.
    - Clicking **Delete** stops calibration and removes the layout.
    - Bass management does not affect noise generation, except for noise on the subwoofer output, which is passed through a lowpass filter on the selected cutoff frequency.
    - Controls and options on all of the pages of the **Preferences** window are disabled. These controls include **Monitor Control**, **Configurations**, **Dolby Headphone**, and **Video**.

- Trim and delay modifications during noise calibration:
  - The modified value is applied live.
  - Switch calibration changes from its automatic mode to its manual mode. In this mode, noise is constantly generated on the speaker where the modification occurs.
  - To switch to another speaker, modify either the **Trim** or **Delay** value for the respective speaker.
  - To return calibration to its automatic mode, stop calibration and start it again.
  - Modified **Trim** or **Delay** values are not saved during calibration. You must manually save the changes by clicking **Save**.

7 Reference information

Dolby Media Decoder leverages particular Dolby technologies.

- Dynamic range control
- Downmixing in Dolby systems
- Dolby Atmos technology
- Dolby TrueHD technology
- Dolby Digital Plus technology
- MLP Lossless technology
- Dolby Digital technology
- Dolby Surround / Dolby Pro Logic technology
- Dolby Pro Logic II technology
- Dolby Surround EX technology
- Dolby Pro Logic IIx technology

7.1 Dynamic range control

Dynamic range control applies to Dolby TrueHD, Dolby Digital Plus, or Dolby Digital bitstreams during decoding.

Decoders interpret dynamic range control metadata to control compression for two-channel downmix overload protection mode and late-night listening modes.

Encoders include dynamic range control metadata in the bitstream to direct the behavior of the compressor in the decoder. The Dolby Media Encoder includes the following dynamic range control presets:

Film Light (default)

Provides moderate compression parameters in the bitstream that the listener may choose to apply during playback.

- Max boost: 6 dB (below –53 dB)
- Boost range: –53 to –41 dB (2:1 ratio)
- Null band width: 20 dB (–41 to –21 dB)
- Early cut range: –21 to –11 dB (2:1 ratio)
- Cut range: –11 to +4 dB (20:1 ratio)

Film Standard

- Max boost: 6 dB (below –43 dB)
- Boost range: –43 to –31 dB (2:1 ratio)
- Null band width: 5 dB (–31 to –26 dB)
- Early cut range: –26 to –16 dB (2:1 ratio)
- Cut range: –16 to +4 dB (20:1 ratio)

Music Standard

- Max boost: 12 dB (below −55 dB)
- Boost range: 55 to −31 dB (2:1 ratio)
- Null band width: 5 dB (−31 to −26 dB)
- Early cut range: −26 to −16 dB (2:1 ratio)
- Cut range: −16 to +4 dB (20:1 ratio)

Music Light

- Max boost: 12 dB (below −65 dB)
- Boost range: −65 to −41 dB (2:1 ratio)
- Null band width: 20 dB (−41 to −21 dB)
- Early cut range: N/A
- Cut range: −21 to +9 dB (2:1 ratio)

Speech

- Max boost: 15 dB (below −50 dB)
- Boost range: −50 to −31 dB (5:1 ratio)
- Null band width: 5 dB (−31 to −26 dB)
- Early cut range: −26 to −16 dB (2:1 ratio)
- Cut range: −16 to +4 dB (20:1 ratio)

Compression choices are available for both Line mode and RF mode. The content producer chooses which of these profiles to assign to each mode. These profiles are applied when the decoder selects a dynamic range control mode.

Dynamic range control profile settings are dependent on an accurate dialogue level setting. Improper setting of the dialogue level parameter may result in excessive and audible application of overload-protection limiting.

The dialogue level value determines the bitstream `dialnorm` parameter, and sets the position of the compressor null band where the signal is neither boosted nor attenuated. The null band is designed to protect audio at the average level of the program from having either cut or boost applied when dynamic range control is applied in the decoder.

Dynamic range control metadata provides graceful control of excessive dynamic range and overload protection. For example, consider a 5.1-channel program with signals at digital full scale on all channels being played through a stereo, downmixed line-level output: without some form of attenuation or limiting, such output signal would certainly clip. Accurately setting the dialogue level and dynamic range control profiles prevents clipping and the unnecessary application of automatic overload protection.

7.2 Downmixing in Dolby systems

Downmixing settings in Dolby Media Producer Master Suite are based on the encoding format, target, and expected listening environments.

Stereo downmixing

The following stereo downmix options are supported:

Dolby TrueHD

Standard: Two-channel stereo Left, Right (Lo/Ro)
Custom: Includes adjustable controls for setting levels

Dolby Digital Plus and Dolby Digital

Surround: Two-channel Left, Right (matrix-encoded Lt/Rt signal)
Stereo: Two-channel stereo Left, Right (Lo/Ro)
Dolby Pro Logic II: Two-channel Left, Right (Dolby Pro Logic II matrix-encoded Lt/Rt signal)

Dolby Digital Plus and Dolby Digital encoding can apply phase 90 filters. The phase 90 filters used by default provide the all-pass 90-degree phase shift filtering for the Ls/Rs feeds into the downmix. This reduces undesirable signal cancellation in stereo presentations, improves imaging, and enables proper matrix decoding. It is strongly recommended to use the 90-degree phase shift for any Lt/Rt downmixes.

Stereo downmixing involves an initial 7.1 to 5.1 downmix, followed by a 5.1 to 2.0 downmix in the same manner as 2.0 consumer products. The coefficients for the two-channel downmixes from 5.1 are as follows:

Lt/Rt (legacy)

Lt = L + (–3 dB x C) – (–3 dB x Ls) – (–3 dB x Rs)
Rt = R + (–3 dB x C) + (–3 dB x Ls) + (–3 dB x Rs)

Lt/Rt (Pro Logic II)

Lt = L + (–3 dB x C) – (–1.2 dB x Ls) – (–6.2 dB x Rs)
Rt = R + (–3 dB x C) + (–6.2 dB x Ls) + (–1.2 dB x Rs)

LoRo

Lo = L + (–3 dB x C) + (–3 dB x Ls)
Ro = R + (–3 dB x C) + (–3 dB x Rs)

7.3 Dolby Atmos technology

The Dolby Atmos experience combines a channel-based audio bed with object-oriented sound to place and move specific effects around the listener.

Using dynamic audio objects, sound designers and artists are free to mix audio in a three-dimensional space, steering effects throughout the room. When a Dolby Atmos soundtrack is played back in the cinema, the object-based audio mix is adapted to the room: audio is rendered to the exact speaker layout of the theatre.

A major benefit of Dolby Atmos is the ability to scale a soundtrack to different room sizes and speaker configurations. Dolby Atmos provides an optimal presentation of the soundtrack for each listening environment.

Audio/video receiver and home-theater-in-a-box products were the first consumer products to support Dolby Atmos. These products decode, scale, and calibrate Dolby Atmos content to accommodate a variety of home theater configurations.

Dolby Atmos for home theater has the following features to produce a three-dimensional audio experience:

- Object-based Dolby Atmos content
- Dolby Atmos decoders
- Dolby advanced audio postprocessing
- Dolby Atmos enabled speaker technology

7.4 Dolby TrueHD technology

Dolby TrueHD is a lossless coding technology for high-definition disc-based media.

Dolby TrueHD delivers sound that is identical to the studio master, unlocking the true high-definition entertainment experience on current and future surround formats on Blu-ray Disc.

Features

- Lossless coding technology.
- Supports carriage of Dolby Atmos content.
- Supports data rates up to 18 Mbps.
- Supports up to eight full-range channels of 24-bit/96 kHz audio and up to six channels of 24-bit/192 kHz audio.
- Supported by HDMI.
- Supports extensive metadata, including dialogue normalization and dynamic range control.
- Supports multiple discrete presentations that can be custom mixes for Dolby Atmos, 7.1, 5.1, or stereo.

Benefits

- Studio master-quality sound that unlocks the true high-definition entertainment experience on next-generation discs.
- Compatible with Dolby Digital audio/video receivers and home-theater-in-a-box products.
- Dialogue normalization that can maintain the same volume level when you change to other Dolby TrueHD , Dolby Digital Plus, and Dolby Digital programming.
- Dynamic range control features enable the customization of audio playback to reduce peak volume levels while experiencing all the details in the soundtrack.
- Selected as a format for Blu-ray Disc.

7.5 Dolby Digital Plus technology

Dolby Digital Plus combines the efficiency to meet future broadcast demands with the power and the flexibility to realize the full audio potential of the high-definition experience.

Built over the foundation of Dolby Digital, Dolby Digital Plus provides even higher quality audio, more channels, and greater flexibility, but remains fully compatible with all current audio/video receivers.

Features

- Multichannel sound with discrete channel output.
- Supports carriage of Dolby Atmos.
- Channel and program extensions can carry multichannel audio programs of up to 7.1 channels and support multiple programs in a single encoded bitstream.
- Outputs a Dolby Digital bitstream for playback on existing Dolby Digital systems.
- Supports data rates as high as 6 Mbps.
- Bit-rate performance of up to 1.6 Mbps on Blu-ray Disc.

- Interactive mixing and streaming capability in advanced systems.
- Supported by HDMI.

Benefits

- Can deliver 7.1 channels or more enhanced-quality audio at up to 6 Mbps.
- Allows multiple languages to be carried in a single bitstream.
- Offers audio professionals new creative power and freedom.
- Compatible with the millions of home entertainment Dolby Digital systems.
- No latency or loss of quality in the conversion process. Maintains high quality at more efficient broadcast bit rates (<192 kbps for 5.1-channel audio).
- Selected by the Advanced Television Systems Committee (ATSC) as the standard for future broadcast applications; named as an option by the Digital Video Broadcasting (DVB) project for satellite and cable TV.
- Selected as a format for Blu-ray Disc.

7.6 MLP Lossless technology

MLP Lossless, licensed by Dolby Laboratories, is the core technology of Advanced Resolution multichannel and stereo DVD-Audio.

MLP Lossless enables producers to encode up to six channels of 24-bit/96 kHz audio or two channels of 24-bit/192 kHz audio onto a DVD-Audio disc, resulting in playback that is bit-for-bit identical to the studio master. Nothing is lost during the encoding and decoding process.

Widely recognized as the first major step forward in high-fidelity audio after the compact disc, DVD-Audio makes your home audio system sound like a live musical performance. DVD-Audio combines DVD technology with MLP Lossless coding to give you music with full 5.1-channel surround sound and fidelity, which is dramatically better than that of the CD. A DVD-Audio player provides an excellent listening experience for DVD-Audio discs, but many DVD-Video player can also play DVD-Audio discs.

7.7 Dolby Digital technology

In thousands of cinemas and millions of homes worldwide, Dolby Digital is the standard for surround sound technology in general and 5.1-channel surround sound in particular.

Dolby Digital is a highly sophisticated and versatile audio encoding/decoding technology. Dolby Digital technology can transmit mono, stereo (two-channel), or up to 5.1-channel surround sound (discrete multichannel audio).

The superior coding efficiency of Dolby Digital, and its ability to deliver high-quality discrete multichannel audio without compromising video quality, has made it the designated audio standard for DVD worldwide.

Dolby Digital is also the preferred multichannel audio standard for broadcast direct to home satellite and digital cable systems. Nearly 60 million digital cable and satellite set-top boxes are currently in use worldwide, supported by over 30 million audio/video receivers equipped with 5.1-channel Dolby Digital decoding. Virtually every DVD-Video player sold today offers a two-channel downmix of the Dolby Digital 5.1-channel movie soundtrack, as well as a digital output for connection to a 5.1-channel audio/video receiver.

In the film industry, the Dolby Digital soundtrack is optically encoded right on the filmstrip, in the space between the sprocket holes. Placing the soundtrack directly on the film allows it to coexist with the analog track, a cost-effective and simple solution for film distributors and theatre owners. The sprocket hole area has also proven highly resistant to wear and tear, so that a soundtrack encoded in Dolby Digital remains free of pops and hiss for the useful life of the print.

The sound information contained in each of the available Dolby Digital channels is distinct and independent. These six channels are described as a 5.1-channel system, because there are five full-bandwidth channels (Left, Right, Center, Left Surround, and Right Surround) with a 3 Hz to 20 kHz frequency range, plus a Low-Frequency Effects (LFE) subwoofer channel devoted to frequencies from 3 to 120 Hz.

7.8 Dolby Surround / Dolby Pro Logic technology

Dolby Surround is the consumer version of the original Dolby multichannel analog film sound formats (Dolby Stereo and Dolby SR).

When a Dolby Surround soundtrack is produced, four channels of audio information (Left, Center, Right, and mono Surround) are matrix encoded onto two audio tracks. These two tracks are then carried on stereo program sources such as stereo soundtracks on DVDs and Blu-ray Disc, videotapes, and TV broadcasts into the home, where they can be decoded by Dolby Pro Logic to re-create the original multichannel surround sound experience.

Dolby Pro Logic was the foundation for multichannel home theater, and was the reference decoder for creating the surround sound audio tracks in thousands of commercially available videocassettes, laser discs, DVDs, and television programs.

With the introduction of the Dolby Digital multichannel film sound format, Dolby Digital replaced Dolby Surround as the preferred technology to deliver multichannel audio to consumers via DVD-Video, digital television, and games. However, every Dolby Digital decoder also provides a stereo signal compatible with Dolby Pro Logic on its analog outputs.

Related information

UPMIXING section on page 31

7.9 Dolby Pro Logic II technology

Dolby Pro Logic II technology processes any high-quality stereo (two-channel) audio into five playback channels of full-bandwidth surround sound.

Dolby Pro Logic II is matrix surround decoding technology that detects the directional cues that occur naturally in stereo content, using these elements to create a five-channel surround sound playback experience. Dolby Pro Logic II is ideally suited for home theater systems, PCs, game consoles, and multichannel in-car audio systems.

Dolby Pro Logic II is fully compatible with all Dolby Pro Logic technologies. It provides optimal audio for playback in a 5.1-channel home theater system for the thousands of videocassettes and TV programs encoded in four-channel Dolby Surround (the encoding counterpart to the Dolby Pro Logic decoding technology).

Dolby Pro Logic II also enables video game consoles to encode five-channel surround sound information into a stereo signal with virtually no impact on the console CPU, which means better audio with no impact on game play.

In addition to enhancing playback, Dolby Pro Logic II can be used to encode TV programming to deliver a surround sound experience for viewers with stereo TV systems.

Related information

UPMIXING section on page 31

7.10 Dolby Surround EX technology

Dolby Surround EX augments the standard 5.1-channel cinema experience with a new Back Surround channel, reproduced by speakers placed directly behind the audience.

The Dolby Surround EX 6.1-channel format added a new dimension to the Dolby Digital 5.1 cinematic experience, and was soon introduced in DVD releases and 6.1 home theater playback equipment as well. The additional channel was matrix-encoded onto the regular Left Surround and Right Surround channels of a conventional Dolby Digital 5.1 soundtrack. This allowed it to be either matrix-decoded by a Dolby Surround EX system, or reproduced by the traditional Left Surround and Right Surround speakers in an older 5.1 system.

An EX flag in the bitstreams allows home theater equipment to detect content with this matrix-encoded Back Surround channel and reproduce it correctly.

The introduction of Dolby Pro Logic IIx 7.1 channel home theater systems surpassed the 6.1 experience and gained rapid adoption. The Dolby Pro Logic IIx system matrix encodes not one but two rear surround signals into the conventional Ls and Rs channels.

The EX flag in the bitstream is re-purposed to indicate presence of any matrix-encoded rear surround signals in a Dolby TrueHD, Dolby Digital Plus, or Dolby Digital 5.1-channel presentation, including the 5.1-channel presentation heard when decoding a 7.1-channel bitstream on a system with only 5.1 capability.

7.11 Dolby Pro Logic IIx technology

Dolby Pro Logic IIx was designed to expand upon several multichannel technologies, including Dolby Pro Logic II matrix decoding as well as Dolby Digital and Dolby Digital EX audio systems by deriving an additional pair of independent rear surround channels.

The additional Dolby Pro Logic IIx channel pair offers decoders additional flexibility in deriving up to 7.1 channels to feed all available speaker outputs of a surround sound system, and to do so from a wide range of sources (from stereo to discrete multichannel).

Whereas 7.1 rear surround channels add an extra dimension to surround sound systems, all Dolby Pro Logic IIx outputs are intended for speaker locations fixed on a horizontal plane.

Features

- Ability to generate up to 7.1 channels from stereo and 5.1 sources.
- Support of several decoding modes:
  - Music mode
  - Movie mode
  - Matrix mode
  - Dolby Digital EX mode
- Support for upmixed output configurations based on stereo sources:
  - 2 to 3 (L and R to L, C, and R)
  - 2 to 4 (L and R to L, R, Ls, and Rs)
  - 2 to 5.1 (L and R to L, C, R, Ls, and Rs)

    - 2 to 6.1 (L and R to L, C, R, Ls, Rs, and Cs)
    - 2 to 7.1 (L and R to L, C, R, Ls, Rs, Lrs, and Rrs)
- Support for upmixed output configurations based on 5.1 sources with Independent Ls and Rs channels:
    - 5.1 to 6.1 (Ls and Rs to Ls, Rs, and Cs)
    - 5.1 to 7.1 (Ls and Rs to Ls, Rs, Lrs, and Rrs)

Note: For 5.1-channel sources, Dolby Pro Logic IIx operates on the Ls and Rs channels only; L, C, R, and LFE channels are bypassed.

Related information

UPMIXING section on page 31

8 Third-party software

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- Disclaimers
- Open-source software

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